

GenAI: Test Cycle Closure

This module explains how **Generative AI assists in the final phase of STLC – Test Cycle Closure**, ensuring proper documentation, metrics, and continuous improvement.

What is Test Cycle Closure?

Test Cycle Closure is the final phase of STLC where teams:

- Evaluate test completion
 - Analyze defects
 - Document lessons learned
 - Prepare test summary reports
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Role of GenAI in Test Cycle Closure

Generative AI helps by:

- Generating test summary reports
 - Analyzing defect trends
 - Creating closure checklists
 - Documenting lessons learned
 - Archiving test artifacts
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How GenAI Supports Test Cycle Closure

Area	GenAI Contribution
Reporting	Auto-generate summaries
Metrics analysis	Highlight trends
Documentation	Create closure docs
Knowledge capture	Lessons learned

Test Cycle Closure: Example

Input Data

- Total test cases: 200
- Passed: 180
- Failed: 15
- Blocked: 5

Prompt

Generate test summary report.

GenAI Output

Test execution completed successfully.
90% pass rate achieved.
Key issues found in payment module.
Recommendation: Improve error handling.

Demo: Automating Test Cycle Closure Using GitHub Copilot

Step 1

Create file `test_closure.md`

```
<!-- Generate test cycle closure report -->
```

Step 2

Copilot Suggests

```
## Test Summary
- Total: 200
- Passed: 180
- Failed: 15
- Blocked: 5

## Risks
- Payment failures

## Recommendations
- Fix payment defects
- Retest
```

Benefits

- Faster reporting
 - Better insights
 - Reduced manual effort
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Limitations

- Needs validation
 - May produce generic summaries
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Quick Check

Q1. What is test cycle closure?

Final STLC phase

Q2. True/False

GenAI can generate test summary reports.

True

Q3. Best practice?

Review AI output

Summary

Generative AI improves Test Cycle Closure by:

Automating reports
Analyzing trends
Capturing knowledge

But **human validation remains essential.**
